



Role of interfaces on phase formation during severe plastic deformation

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ABSTRACT

The role of interfaces on phase formation during severe plastic deformation (SPD) was investigated using molecular dynamics simulations. It is found that dilute solute additions that segregate to interphase boundaries, and strengthen them, can greatly extend the solubility limits of the bounding phases at steady state, providing a novel approach to processing nanocomposite alloys by SPD. The results are rationalized in terms of a modified effective temperature model, whereby increasing the interface strength leads to higher effective temperatures and correspondingly to higher solubilities. Comparison of the segregation coefficient at the interphase boundary with solubilities in the bounding phases illustrates the effective temperature model is self-consistent and comprehensive. Past experimental studies on severe plastic deformation are found consistent with these observations.

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1. Introduction

It is well recognized that severe plastic deformation (SPD) can greatly extend solubility limits in nominally two-phase alloys [1–6] and in many cases produce solid solutions [7–11]. Two important materials parameters often discussed for controlling forced chemical mixing during SPD are the heat of mixing of the alloy and the elastic properties of the coexisting phases [12–18]. In addition to these mechano-chemical properties, there is evidence that the interface properties of coexisting phases can also play a significant role in the microstructural evolution during SPD [17,19–24], although much of this work has focused on specific structural interfaces and at low strains. At low strains, however, typically < 10 – 100 , the predominant microstructural changes during SPD are kinetic roughening and grain refinement, and not chemical mixing [25–29]; the crossover between these two regimes, however, is gradual, and it depends on the length scale of the system [28] see, however, refs. [30,31] for details on morphology evolution in immiscible Cu-based alloys over a wide range of strains. The low-strain regime is particularly important, for example, during sliding wear since the near surface region “wears” away before reaching high strains [26,32–34]. The high strain regime refers to systems that have undergone prolonged plastic deformation, and phase evolution approaches a steady state, predominantly through forced chemical mixing processes. We show here for this high

strain regime, which is relevant for processing bulk nanocomposite materials, for example by ball milling [35], accumulative roll bonding [36], or high-pressure torsion (HPT) [37], that the strength of interphase boundaries plays a key role in determining steady-state phases during SPD.

One of the noteworthy features of the steady-state regime is that the alloy phases that coexist, and their compositions, are generally independent of the initial state of the alloy system. Martin has addressed such behavior in driven alloys by deriving an *effective potential*, which plays the same role in driven systems as the free energy does in systems near equilibrium [38,39]; this formalism thus explains why systems adopt a unique phase structure in steady state. For equilibrium systems described by a regular solution model, moreover, Martin further showed that the effective potential leads to the rule of corresponding states, whereby the phases present in a system driven at temperature T are determined by the equilibrium phase diagram, but at some higher temperature, T_{eff} , where $T_{eff} = T(1 + \gamma_b)$. Here, $\gamma_b = D_b/D_{th}$, where D_{th} is the thermally activated interdiffusion coefficient at temperature, T , and D_b represents interdiffusion by “ballistic” forced mixing of atoms. In this model, it is assumed that D_b is independent of chemical interactions and that the ballistic flux obeys Fick’s law, i.e., solute diffusion is down gradients in the concentration, as if the system were at an infinite temperature. While this effective temperature model works well for alloys with moderate chemical interactions between alloy components and at shearing temperatures where thermally activated diffusion processes take place [1,40–44], it breaks down for highly reactive and strongly immiscible alloys,

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as the chemical driving forces begin to compete with the external forcing. The model has since been modified to account for the effects of these stronger chemical interactions on ballistic mixing [45,46], yielding, in the case of ref. [46]

$$T_{eff} = T(1 + \gamma_b^{T_0}) / (1 + \frac{T}{T_0} \gamma_b^{T_0}), \quad (1)$$

where, $\gamma_b^{T_0} = D_b^{T_0} / D_{th}$; D_{th} is again the thermally activated interdiffusion coefficient at temperature T , $D_b^{T_0}$ is the interdiffusion coefficient for forcing mixing, but now operating as if the sample were at some finite temperature T_0 , the so-called “process temperature” [46], and the ballistic flux is now down the gradient in the chemical potential, also evaluated at T_0 . It is seen from Eq. (1) that at low temperatures, for which D_{th} is small, $T_{eff} \rightarrow T_0$; this typically occurs below room temperature, as vacancies then become immobile and fast diffusion along grain boundaries and dislocations is suppressed [47,48]. The present work considers this low temperature regime, and T_{eff} and T_0 will be used interchangeably.

MD simulations have proved an extremely valuable tool for studying alloys subjected to external driving forces in alloys, for example energetic particle irradiation [49–51], wear [32,52,53], and SPD [14,54–56]. This is particularly true at low temperatures where thermally activated diffusion can be neglected so that the external forcing can be greatly accelerated. For low-temperature SPD, for example, strains > 1000 are readily achieved in systems containing over 10^5 atoms. In regard to the effective temperature model, past MD studies of SPD on highly immiscible, two-phase alloys have illustrated that the process temperature T_0 in Eq. (1) scales with the shear modulus of the alloy [14], and that the steady-state solubility and chemical short-range order scale with the heat of mixing [14,45]. These past findings, however, neglect any explicit role of interfaces on phase formation; we show here that the strength of interfaces, in fact, can play a dominant role in determining the stability of multi-phase alloys under steady state SPD, and particularly so in nanocomposite materials. We illustrate this behavior by employing MD simulations of SPD to examine phase stability in highly immiscible alloys which contain dilute ternary alloy additions that segregate to the phase boundaries during SPD and alter the interface properties. The new understanding gained from this work provides a comprehensive and self-consistent description of phase coexistence during SPD of highly immiscible alloys and strongly reacting systems within the effective temperature model, and it suggests how dilute solute additions can provide a useful control variable for tailoring phase stability in nanocomposite systems.

2. Computational Methods

The systems investigated here are represented by three ternary pseudo-alloys of $\text{Cu}(1)_{(50-x)}\text{Cu}(2)_{(50-x)}\text{Cu}^*_{2x}$, $\text{Cu}_{(50-x)}\text{Co}_{(50-x)}\text{Cu}^*_{2x}$, and $\text{Cu}_{(50-x)}\text{Co}_{(50-x)}\text{Co}^*_{2x}$. In elemental form, the three types of Cu, i.e., Cu(1), Cu(2) and Cu^* , and two types of Co, Co and Co^* are described by Cleri-Rosato type interatomic potentials derived for Cu and FCC-Co in refs. [57] and [58], respectively. The cohesive energy in this formulation is given by

$$E_c = \sum_{i=1}^N \left[\sum_{j=1}^N A_{\alpha\beta} \exp \left[-p_{\alpha\beta} \left(-\frac{r_{ij}}{r_{\alpha\beta}^0} - 1 \right) \right] - \sqrt{\sum_{j=1}^N \xi_{\alpha\beta}^2 \exp \left[-2q_{\alpha\beta} \left(-\frac{r_{ij}}{r_{\alpha\beta}^0} - 1 \right) \right]} \right]$$

where α and β refer to the chemical identity of atoms i and j , and $r_{\alpha\beta}^0$ is the nearest neighbor distance separating α and β atoms. Parameter values for the metals of interest here are provided in the Supplementary Information (see Table S1) as well as in refs. [57] and [58]. The advantage of this formulation, as shown previously, is that by varying the parameter $\xi_{\alpha\beta}$ the heat of mixing between any two alloy components, for example Cu(1) and Cu(2),

can be systematically varied, but with little effect on the elastic properties of the alloy [14,56]. In the present work $\xi_{\alpha\beta}$ was selected to yield a very high heat of mixing of approximately 2 eV/atom between the principal alloying components at infinite dilution, while the dilute ternary component, Cu^* or Co^* , was given a near-zero heat of mixing with respect to each of the principal alloying components. The very large heat of mixing is necessary for the alloys to remain two-phase during extended low-temperature shearing, while the near-zero heat of mixing for the dilute ternary component results in their segregation to the interphase boundary. The $\text{Cu}(1)_{(50-x)}\text{Cu}(2)_{(50-x)}\text{Cu}^*_{2x}$ system, therefore, enables study of the effects of Cu^* segregation on phase stability during SPD with minimal influence of the elastic properties of the two co-existing phases, whereas the $\text{Cu}_{(50-x)}\text{Co}_{(50-x)}\text{Cu}^*_{2x}$ system explores the combined effects of segregation and the relative elastic properties of the two phases. We selected FCC cobalt for this study since the atomic radius of Co is close to that of Cu, thus avoiding issues of amorphization, which have been reported for simulations of SPD in other systems with large size mismatch and either large positive heats of mixing: Cu-Nb, Cu-V, and Cu-Ta [59], or large negative heats of mixing: Cu-Zr [54,60]. Because of the artificially imposed high heat of mixing in Cu-Co alloy, however, this pseudo-alloy does not represent real Cu-Co, which has been well studied experimentally [9,31,61].

The simulations of plastic deformation were performed using LAMMPS [62], with visualizations provided by OVITO [63]. The systems were all FCC containing 256,000 atoms, and they were initiated as bilayers, e.g., a $\text{Cu}(1)_{(100-x)}\text{Cu}^*_x$ α -phase and a $\text{Cu}(2)_{(100-x)}\text{Cu}^*_x$ α' -phase, with a coherent (200) interface that was parallel to one of the cubic, periodic boundaries. The ternary component, Cu^* or Co^* , was initially dispersed randomly throughout the cell. The deformation was applied at 100 K as cyclical, bi-axial compressions along two prescribed principal axes with a total strain rate of $\dot{\epsilon} = 5 \times 10^9 \text{ s}^{-1}$. Our three-step bi-axial compression cycle can be described by the ‘free’ axis, for which $\sigma_{ii} = 0$, following the path $\{y, x, z\}$, with a strain of $\sim 20\%$ per step along each axis. Strain rates along each axis were adjusted up to 2% in order to preserve the shape of the computational cell at the end of each cycle. A schematic of the strain path is shown in the Supplementary Information, Fig. S2. The use of this three-step biaxial stress cycle avoids strain localization. Varying the strain rate from $1 \times 10^8 \text{ s}^{-1}$ to $5 \times 10^9 \text{ s}^{-1}$, moreover, had no significant effect on the results, largely because thermally activated atomic motion is highly suppressed at 100 K.

The interface atoms in the sheared alloys were identified as follows. For $\text{Cu}(1)_{(50-x)}\text{Cu}(2)_{(50-x)}\text{Cu}^*_{2x}$, the Cu(1)-rich and Cu(2)-rich phases, α and α' , respectively, were first obtained using the condition that the fraction of nearest like-atom neighbors within the respective phases be > 0.55 . The interface region was then created by including (i) all atoms within 0.4 nm of the α -phase, but not in the α -phase, plus (ii) all atoms within 0.4 nm of the α' -phase, but not in the α' -phase, making the interface ~ 0.8 nm wide. For a detailed description of this scheme, please refer to Fig. S3, in the Supplementary Information. The topography of the interface region in $\text{Cu}(1)_{47.5}\text{Cu}(2)_{47.5}\text{Cu}^*_5$ during steady state SPD is illustrated in Fig. 1(a), while concentration profiles of the three alloy components across the interface, are plotted in Fig. 1(b). Note that the concentration profiles refer to the average of the profile across the local interface and not to the profile across the average interface position, thus largely avoiding contributions from roughness. Cu(1) and Cu(2) have error function-like profiles, while that for Cu^* is approximately Gaussian. The average concentration of Cu^* within the interface region, $X_{\text{Cu}^*}^{\text{int}}$, is defined here as its average value within the full width at half maximum of the Cu^* distribution. This concentration will be used in what follows as a convenient characterization of the interface, but not for detailed analysis.

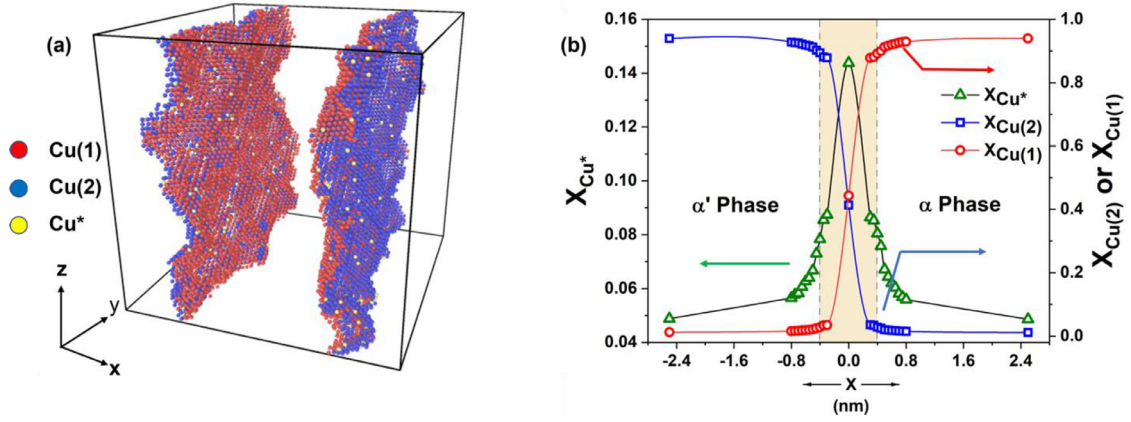


Fig. 1. Interfacial region for $\text{Cu}(1)_{47.5}\text{Cu}(2)_{47.5}\text{Cu}^*_{5}$: (a) Topology of the steady-state interface after SPD; (b) Concentration profiles of Cu(1), Cu(2) and Cu*, $X_{\text{Cu}(1)}$, $X_{\text{Cu}(2)}$, and X_{Cu^*} , respectively, across the interface region (highlighted in yellow).

3. Results and Discussion

3.1. $\text{Cu}(1)_{(50-x)}\text{Cu}(2)_{(50-x)}\text{Cu}^*_{2x}$

The primary results of this work are shown in Fig. 2(a) for the $\text{Cu}(1)_{(50-x)}\text{Cu}(2)_{(50-x)}\text{Cu}^*_{2x}$ alloys, where the concentration of Cu(2) in the Cu(1)-rich α phase, $X_{\text{Cu}(2)}^{\alpha}$, and the concentrations of Cu* in α and the α - α' interface, $X_{\text{Cu}^*}^{\alpha}$ and $X_{\text{Cu}^*}^{\text{int}}$, respectively, are plotted as functions of strain for average Cu* (nominal) concentrations between 0 and 0.10. It is observed that $X_{\text{Cu}(2)}^{\alpha}$ and $X_{\text{Cu}^*}^{\text{int}}$ both increase initially with strain, but then approach constant values after a strain of ~ 200 . The characteristic strain to reach steady state, as noted above, will depend on the microstructural length scale of the system, e.g., the precipitate size and density [64,65], or in the current study, the size of the computational cell. Values for the steady-state concentrations of Cu(2) in α , $X_{\text{Cu}(2)}^{\alpha(\text{SS})}$, and Cu* in the interface, $X_{\text{Cu}^*}^{\text{int}(\text{SS})}$, are plotted in Fig. 2(b) as a function of the nominal Cu* concentration. It is observed that Cu* indeed segregates to the α - α' interface during low-temperature plastic deformation and that both $X_{\text{Cu}(2)}^{\alpha(\text{SS})}$ and $X_{\text{Cu}^*}^{\text{int}(\text{SS})}$ increase dramatically with increasing concentration of Cu*. Notably, the segregation occurs in complete absence of thermally-activated diffusion, illustrating that chemical interactions can indeed influence ballistic mixing, as previously reported [45,46,66]. The process temperature, T_0 defined in Eq. (1) can be deduced from the data in Fig. 2 by locating the temperature in the equilibrium phase diagram at which the sol-

ubility equals the steady-state concentrations during SPD. For the low solubilities of these highly immiscible alloys, the regular solution model provides a reasonable description of the alloy, as will be discussed below, and thus T_0 can be approximated by,

$$kT_0 \approx -\Delta H_{\text{mix}} / \ln X_{\text{Cu}(2)}^{\alpha} \quad (2)$$

where ΔH_{mix} has been adjusted slightly in Eq. (2) to account for the small concentrations of Cu* in the α and α' phases [67].

The results for T_0 are plotted in Fig. 3 as a function of $X_{\text{Cu}^*}^{\text{int}(\text{SS})}$. The value of T_0 is seen to increase from $\sim 3,600$ K for $X_{\text{Cu}^*}^{\text{int}(\text{SS})} = 0$ to $\sim 6,300$ K for $X_{\text{Cu}^*}^{\text{int}(\text{SS})} = 0.15$.

The above results can be understood within the framework of the modified effective temperature model, by first recalling from above that T_0 scales with the shear modulus of the material [14]. This finding can be rationalized by noting that the average work done on an atom during plastic deformation, $\sim \sigma_{\text{max}} \cdot \epsilon \cdot \Omega_0$, scales with the shear modulus [68]; here σ_{max} is the shear strength of the material, ϵ is the strain, and Ω_0 is the atomic volume. For example, assuming σ_{max} is on the order of the theoretical strength, σ_{th} , and $\sigma_{\text{th}} \sim G/10$ (where G is the shear modulus) [69], then in Cu, $\sigma_{\text{max}} \approx 4.5$ GPa. (For comparison, the flow stress measured in the current simulations is ~ 3.5 GPa.) Accordingly, the average work done on each atom undergoing one jump during shear, i.e., after a local strain of unity, is ~ 0.35 eV, yielding T_0 on the order of 4,000 K. While the above picture is very qualitative, this discussion is nonetheless relevant here since an increasing body of experimental and simulation work has demonstrated that nanocrystalline metals

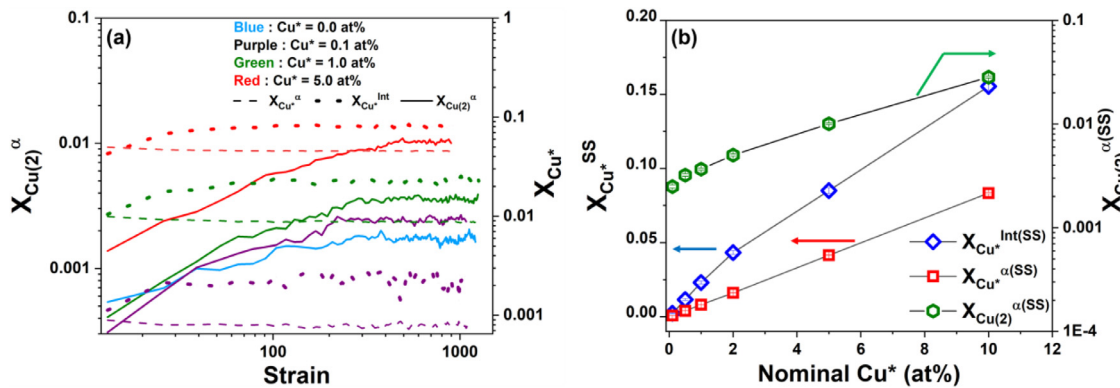


Fig. 2. (a) Concentrations $X_{\text{Cu}(2)}^{\alpha}$, $X_{\text{Cu}^*}^{\alpha}$ and $X_{\text{Cu}^*}^{\text{int}}$ as a function of strain during SPD for representative alloys: $X_{\text{Cu}^*}^{\alpha}$ (---), $X_{\text{Cu}^*}^{\text{int}}$ (---), $X_{\text{Cu}(2)}^{\alpha}$ (—). Colors red, green, purple, and blue represent different nominal concentrations of Cu*. (b) Steady-state concentrations $X_{\text{Cu}(2)}^{\alpha(\text{SS})}$, $X_{\text{Cu}^*}^{\alpha(\text{SS})}$, and $X_{\text{Cu}^*}^{\text{int}(\text{SS})}$ as a function of Cu* concentration: margin of error is $\sim 6.0\%$.

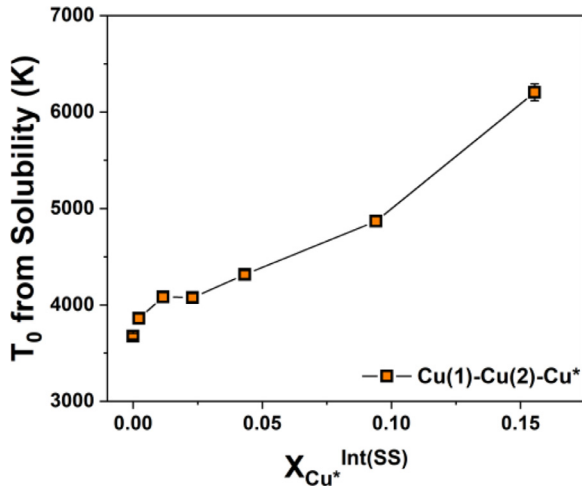


Fig. 3. Effective temperature (T_0) as a function of $X_{Cu^*}^{Int(SS)}$, as derived from the regular solution model, see Eq. (2).

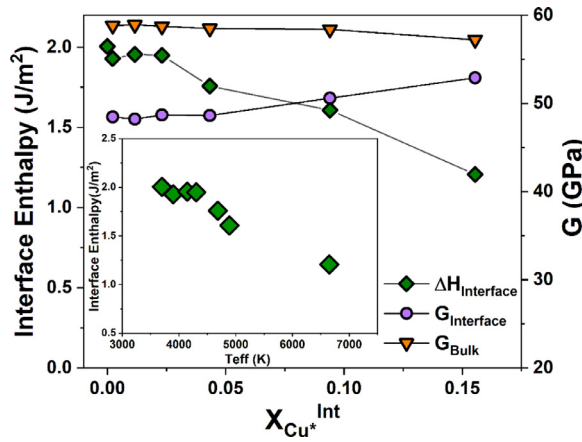


Fig. 4. Interface free energy and local shear moduli as a function of $X_{Cu^*}^{Int(SS)}$. INSET: Interface enthalpy (ΔH_{int}) as a function of the effective temperature (T_0). T_0 was obtained from Fig. 3.

can be greatly strengthened by doping the grain boundaries with appropriate solute atoms [70–75], specifically those solutes that lower grain boundary energies [72]. This strengthening should thus raise the effective temperature, or T_0 . While the α – α' interface represents a coherent interphase boundary, and not a grain boundary, we can expect similar strengthening behavior and an increase in T_0 in this system by doping the interphase boundary with Cu^* , which lowers the interface energy, γ_{IE} . Fig. 4, for example, shows the reduction in interfacial enthalpy, ΔH_{int} , (the main contribution to γ_{IE} in these highly immiscible alloys), as a function $X_{Cu^*}^{Int(SS)}$, as well as changes in the local shear modulus, calculated for the bulk phases α and α' and for the interfacial region. The local shear moduli were evaluated using a modified version of LAMMPS, based on the method detailed in ref. [76], while ΔH_{int} was obtained by employing LAMMPS to determine the interface enthalpy directly. The shear modulus is seen in Fig. 4 to decrease slightly in the bulk phases with increasing $X_{Cu^*}^{Int(SS)}$. This is due, presumably, to the small increase in the steady-state solubilities of the two bulk phases and the creation of high energy Cu(1)–Cu(2) bonds. The interface region, in contrast, gains Cu^* atoms, thereby lowering the number of Cu(1)–Cu(2) bonds, reducing the interface energy, and increasing the local shear modulus. This strengthening thus leads to an increased effective temperature at the interface, and thus enabling Cu(1) atoms to transfer across the interface into Cu(2) and

vice versa. It should be noted that the effective temperature of single crystalline Cu(1) (or Cu(2)), in absence of a Cu(1)–Cu(2) interphase boundary is higher at the interface, reported in ref. [14] to be 7,800 K. This is significant since in asymmetric systems with very large differences in hardness, the effective temperature of the soft phase may be lower than that of the interface, and thus it may control the steady state phases. See e.g., ref [15].

We checked the self-consistency of this picture by also determining the effective temperature of the interface using the segregation coefficient of Cu^* , i.e., by the Langmuir–McLean isotherm. This was carried out here using a canonical Monte Carlo (MC)–MD hybrid model, which is implemented in LAMMPS [77]. This method swaps pairs of Cu(i)–Cu(j) atoms ($i \neq j$) according to a Metropolis scheme, while structurally relaxing the system using MD after a given number of swaps. We employed this model by starting with the states of the various alloys during steady-state SPD and setting all applied stresses to zero. The MC–MD code was then employed to find the temperature that retains the same alloy concentration profiles across the interface; no additional plastic deformation is applied during this procedure. The results are shown in Fig. 5, starting with the evolution of $X_{Cu^*}^{Int}$ as a function of simulation time for different trial values of T_0 , see Fig. 5(a). When the trial value of T_0 is too large, $X_{Cu^*}^{Int}$ decreases, and when too small, it increases. Shown in Fig. 5(b) are the alloy concentration profiles (i) after reaching steady state SPD and (ii) following application of the MD–MC code at the deduced value of T_0 . Concentration profiles for all alloy components are seen to remain nearly unchanged by MC–MD for $T_0 = 5,000$ K. The key findings of this analysis are summarized in Fig. 5(c), where the effective temperatures deduced from segregation are plotted against the effective temperatures determined from the steady state solubilities. The agreement between two values of T_0 is quite good, again indicating that the effective temperature at the interface controls the solubilities in the bounding phases. As a technical aside, we note that the values of T_0 in Fig. 5(c) obtained using the regular solution model, Eq. (2), agree quite well with those obtained using the MD–MC code, as seen in Fig. 5(c).

3.2. $Cu_{(50-x)}Co_{(50-x)}Cu^*_{2x} / Cu_{(50-x)}Co_{(50-x)}Co^*_{2x}$

We also considered forced chemical mixing in a less symmetric system, Cu–Co(FCC), for which the elastic properties of the α and α' phases are no longer identical. Again, the heat of mixing parameter was set close to 2 eV/atom for the two principal alloying elements, Co in the Cu-rich α -phase and Cu in the Co-rich α' -phase, using an appropriate choice of $\xi_{\alpha\beta}$ (See Table S1). A dilute ternary component, either Cu^* or Co^* , was also added to the system and given a near-zero heat of mixing with respect to the two principal alloying elements. The results are presented in Fig. 6 for $Cu_{(50-x)}Co_{(50-x)}Cu^*_{2x}$, where the steady-state concentrations of the three alloy components are plotted as a function of nominal Cu^* concentration; substituting Co^* for Cu^* in $Cu_{49}Co_{49}Cu^*_2$ led to nearly identical results. The primary result observed in Fig. 6 is that SPD behavior in $Cu_{(50-x)}Co_{(50-x)}Cu^*_{2x}$ is very similar to that found in $Cu(1)_{(50-x)}Cu(2)_{(50-x)}Cu^*_{2x}$ system, see Fig. 2(b), i.e., as the average Cu^* concentration increases, Cu^* increasingly segregates to the interphase boundary, and the solubilities $X_{Co}^{\alpha(SS)}$ and $X_{Cu}^{\alpha'(SS)}$ increase, indicative of an increasing effective temperature. In Fig. 7, we illustrate that also like Cu(1)–Cu(2)– Cu^* , this increase correlates with a reduction in the enthalpy of interphase boundary and increase in the local shear modulus [76] in this region. Here local values of ΔH_{int} and G are plotted across the interface to better show the asymmetry.

The Cu–Co– Cu^* results also show differences from those of $Cu(1)_{(50-x)}Cu(2)_{(50-x)}Cu^*_{2x}$, most notably, $X_{Cu}^{\alpha(SS)} \neq X_{Co}^{\alpha(SS)}$, with

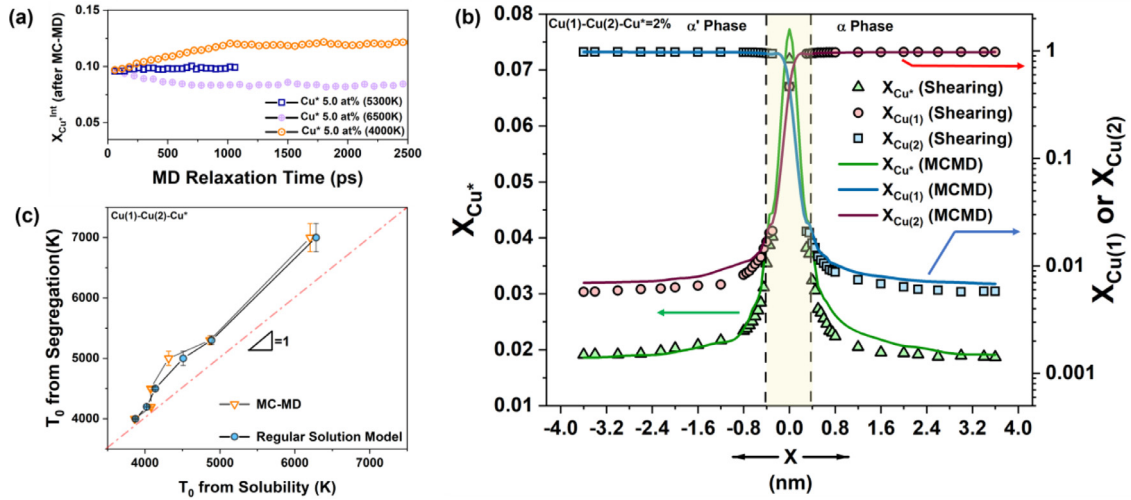


Fig. 5. (a) Concentrations of Cu^* in the interface in $Cu(1)_{47.5}Cu(2)_{47.5}Cu^*_5$ as a function of MD relaxation time (including MC swaps) for different trial values of T_0 . (b) Concentration profiles for all alloy components in $Cu(1)_{49}Cu(2)_{49}Cu^*_2$, showing similar molar concentration profiles during (i) steady state SPD and (ii) following MC-MD at 5000 K. (c) Effective temperature T_0 derived from segregation (using MC-MD) as a function of T_0 determined by the steady-state solubilities, using either the regular solution model or MC-MD.

Table 1

Relative solute concentrations during steady-state SPD and steady-state MC/MD. Also listed are the effective temperatures, T_0 , in the different alloys during SPD. Values of T_0 were estimated using MC/MD fitting of concentrations in each phase, separately. Values deduced from the regular solution model are in parentheses.

	$Cu_{50}Co_{50}$	$Cu_{49.75}Co_{49.75}Cu^*_{0.5}$	$Cu_{49.5}Co_{49.5}Cu^*_{1}$	$Cu_{49}Co_{49}Cu^*_{2}$	$Cu_{48}Co_{48}Cu^*_{4}$
$\Delta X_{Cu^*}^{\alpha'}/X_{ave}$ (SPD)	1.08	1.00	1.03	1.00	0.91
$\Delta X_{solute}^{\alpha'}/X_{ave}$ (MC/MD) α' -best fit	0.266	0.0306	0.03	0.03	0.02
$\Delta X_{solute}^{\alpha'}/X_{ave}$ (MC/MD) α' -best fit	0.493	0.039	0.03	0.06	0.05
T_0^{α} (K)	5,800 (5,900)	5,850 (6080)	5,950 (6,290)	6,100 (6,600)	6,300 (7,210)
$T_0^{\alpha'}$ (K)	4,600 (4,400)	4,700 (4,750)	4,800 (4,830)	5,000 (5,020)	5,200 (5,530)

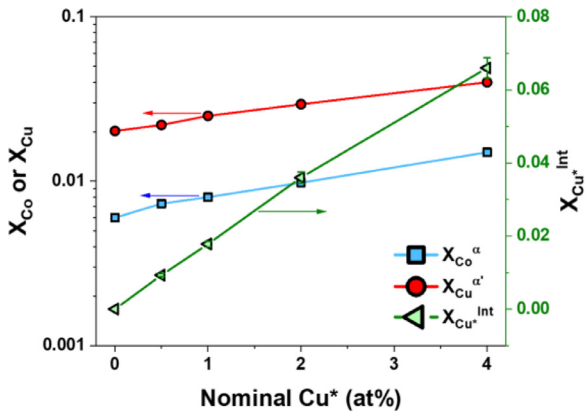


Fig. 6. Steady-state concentrations of Co in α , Cu in α' , and Cu^* at the interface plotted as a function of nominal Cu^* . Error margins are within $\sim 6.0\%$.

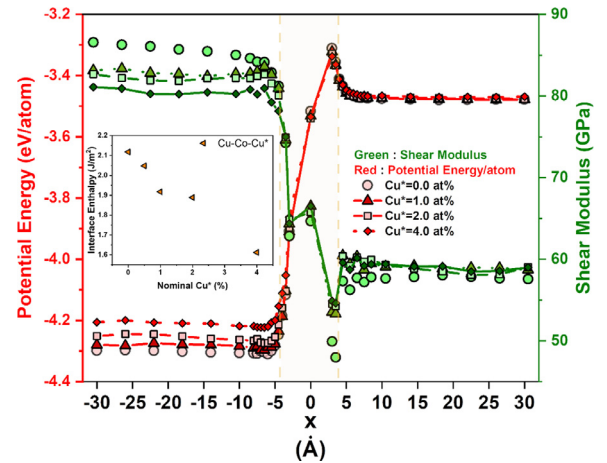


Fig. 7. Local shear modulus profile across the interface (Green). Local average potential energy per atom profile across the interface (Red). (Inset) Average interface enthalpy as a function of the nominal Cu^* concentration. The α' -phase corresponds to the phase on the left.

$X_{Cu}^{\alpha'(SS)}$ being greater than $X_{Co}^{\alpha(SS)}$ for all values of Cu^* . This asymmetry is seen more directly in Fig. 8 where the concentration profiles of $X_{Co}^{\alpha(SS)}$ and $X_{Cu}^{\alpha'(SS)}$ are plotted across the α - α' interface for the binary Cu-Co and ternary $Cu_{49}Co_{49}Cu^*_2$ alloys. We examine this asymmetry by first applying MC/MD, as described above for $Cu(1)$ - $Cu(2)$ - Cu^* ; the results are included in Fig. 8. For both the binary and ternary alloys, it is observed that no single trial value of T_0 reproduces $X_{Co}^{\alpha(SS)}$ and $X_{Cu}^{\alpha'(SS)}$; rather two values of T_0 are required. Since presently we can only apply MC/MD using a single value of T_0 , we assume for now that MC/MD yields approximately the correct values of T_0 in each phase, i.e., for the binary alloy, $T_0 \sim 4,600$ K in α and $\sim 5,800$ K in α' , while for the 2 at.% Cu^* alloy shown in Fig. 8(b) $T_0 \sim 5,000$ K in α and $\sim 6,000$

K in α' . See Table 1 for values of T_0 in the other alloys. As seen in Table 1, the regular solution model for each phase yields similar results, at least until the solute concentrations become too high. It is noteworthy, that MC/MD does show some asymmetry in the concentration profiles for Cu and Co, see Fig. 8, but it is small compared to that following steady state SPD. The asymmetry in these MC/MD profiles is due to interactions between solute and microstructural defects such as dislocations. (see Supplementary Information Fig. S4 and Table S2). Table 1 compares values of $\Delta X_{solute}/X_{ave}$ ($\Delta X_{solute} = (X_{Cu}^{\alpha'} - X_{Co}^{\alpha}); X_{ave} = (X_{Co}^{\alpha} + X_{Cu}^{\alpha'})/2$) ob-

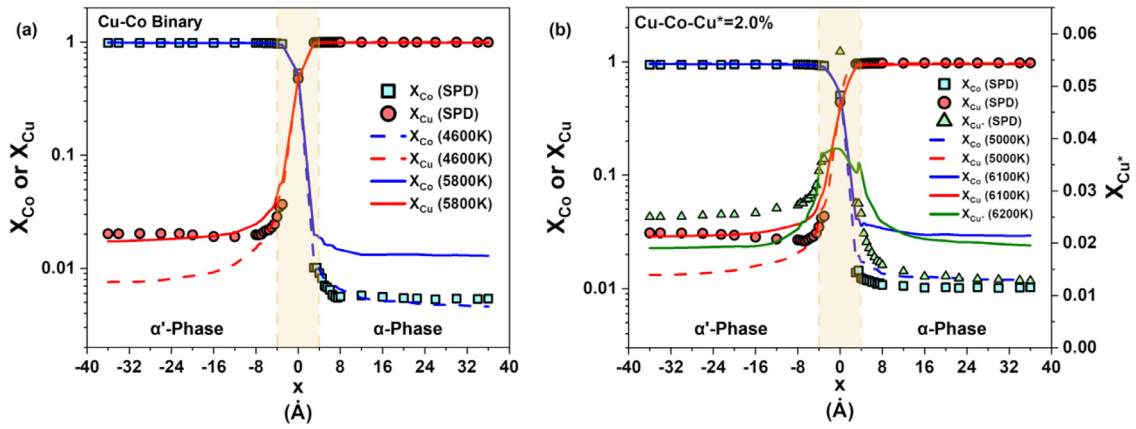


Fig. 8. Steady-state Composition profiles of (a) Cu-Co binary alloy, (b) $\text{Cu}_{49}\text{Co}_{49}\text{Cu}^*_2$ after shearing and MC/MD.

Table 2

(Cu-Co) and (Co-Co) cluster size analysis in α and α' phases for $\text{Cu}_{49}\text{Co}_{49}\text{Cu}^*_2$.

	Total No. of atoms	No. of Co atoms	No. of Cu atoms	Total no. of clusters	Cluster Size = 1 (conc)	Cluster Size = 2 (conc)
α -phase	117363	714	116649	642	590(5.02×10^{-3})	41(3.49×10^{-4})
α' -phase	114075	111750	2325	1838	1517(1.33×10^{-2})	221(1.94×10^{-3})

tained during steady state shearing with those found by application of MC/MD.

The differences in the effective temperatures in the two phases approximated from the solubilities can be corroborated by other alloy properties. For example, solute clustering, or chemical ordering also depends on the effective temperature. The binding energies, E_B , between two Cu atoms in α' and two Co atoms in α in our Cu-Co pseudo-alloy are large, $\sim \frac{\Delta H_m}{6} = 0.33$ eV, (the calculated values are ~ 0.335 eV and ~ 0.366 eV, respectively). Consequently, despite the high effective temperatures and low solute concentrations, clustering is not negligible. As shown in Table 2 for the binary alloy Cu-Co, Cu clusters of size $N = 2$ comprise $\sim 30\%$ of the solute in the α' phase, while Co clusters of size $N = 2$ comprise $\sim 14\%$ of the Co atoms in the α phase. We can estimate the effective temperatures in α and α' phases using the expression $X_{2M} = 6(X_M)^2 \exp(E_B/kT_0)$, where M represents Cu or Co in α' or α , respectively. The results yield $T_0 = 5,100$ K and $6,400$ K in α and α' , respectively, in good agreement with the effective temperatures approximated from the solubility data using MC/MD.

The difference in effective temperatures can similarly be deduced from the relative concentrations of Cu^* in the α and α' phases by considering the chemical potential of Cu^* . Since Cu^* has nearly zero heat of mixing with the principal alloy components, and the alloy concentrations are low in both phases, we approximate the chemical potential by $\mu_{\text{Cu}^*} = RT \ln X_{\text{Cu}^*}$. In steady state, $RT_0^\alpha \ln X_{\text{Cu}^*}^\alpha = RT_0^{\alpha'} \ln X_{\text{Cu}^*}^{\alpha'}$, or $T_0^\alpha / T_0^{\alpha'} = \ln X_{\text{Cu}^*}^{\alpha'} / \ln X_{\text{Cu}^*}^\alpha$. For the $\text{Cu}_{49}\text{Co}_{49}\text{Cu}^*_2$ alloy, shown in Fig. 8(b), we obtain $T_0^\alpha / T_0^{\alpha'} \sim 0.78$, again in good agreement the effective temperatures obtained from the solubility data in Fig. 7(b), $T_0^\alpha / T_0^{\alpha'} \sim 0.82$. We thus conclude from the solubilities of Cu, Co and Cu^* , and solute clustering that the effective temperatures in the α and α' phases are indeed different, with $T_0^{\alpha'} > T_0^\alpha$.

The possibility for different effective temperatures across an interphase boundary during SPD raises fundamental questions about the role of the interfaces within the effective temperature model. Before discussing this issue, however, we emphasize two points: (i) similar to Cu(1)-Cu(2)- Cu^* , segregation of Cu^* (or Co^*) to the Cu-Co interface strengthens the interface and leads to higher solubilities (or effective temperatures) in both phases, in support of the main conclusion of this work; (ii) the difference in the effective temperatures of Cu-rich α and Co-rich α' phases is $\Delta T_0 \sim 1,000$ K,

is relatively small compared to the difference in effective temperatures in the binary pseudo-alloys Cu(1)-Cu(2) and Co(1)-Co(2). For Cu(1)-Cu(2), $T_0 \sim 3,600$ K and Co(1)-Co(2), $T_0 \sim 9,000$ K [27], i.e., $\Delta T_0 \sim 5,400$ K. Thus, the interface is still playing a key role in determining phase-coexistence and solubilities during SPD. While we can only speculate, at present, about the reason for different effective temperatures on opposite sides of an interphase boundary, we suggest two possibilities. The first derives from considerations of alloys under irradiation, a driven system that shares many similarities to SPD. In this case forced chemical mixing results from atoms being displaced a distance R in energetic recoil events [78], where R can exceed a few atomic distances. Thus, the diffusion is nonlocal, and recoil atoms may not fully equilibrate with the effective temperature at a sharp interface as they recoil from one phase to another. Indeed, recent phase field modeling of immiscible alloys at elevated temperatures has shown that in systems containing grain boundaries, the effective temperature model works very well when R is smaller than the grain boundary width, w_{GB} , but it begins to break down as R becomes greater than w_{GB} [79]. For severe plastic deformation, we can thus imagine that the atomic trajectories of atoms through the interface during forced mixing, similar to recoil mixing, influence the diffusion process and thus the apparent effective temperature of the interface, i.e., treating D_B in eq.(1) as a local diffusion coefficient is an approximation in the present model. Since D_B is nonlocal, it will unlikely be symmetric for systems with very different elastic properties, such as Cu and Co. The second possibility concerns the relative strengths of the interface and the softer alloy phase, here, the Cu-rich α -phase. As illustrated in Fig. 7, the shear modulus in the interface region is, on average, larger than in the α -phase (but not α'), and only very close to the α -phase is it lower. This possibly reduces the influence of the interface on α , while still strongly affecting α' , rationalizing why T_0 in the Co-rich α' -phase is reduced from 9,000 K to 6,000 K due to the interface, but it only increases in the Cu-rich α -phase from 3,600 K to 4,600 K.

While the present results show that the steady-state concentrations are determined primarily by the effective temperature, or strength, at the interface and not directly by the properties of the bounding phases, the kinetics of SPD can be very different in these two phases. In this respect, it should be pointed out that in the current simulations the strain path using periodic boundary con-

ditions requires that dislocations glide through both the α and α' phases. In some situations, such as ball milling of W and Al powders [80], it is possible that deformation takes place almost entirely in Al, so that the system may never reach steady state on laboratory time scales. Additional discussion regarding mechanisms of chemical mixing in dissimilar phases can be found in refs. [15–17].

Few experimental results are currently available that directly test this new description of SPD, however, we note high pressure torsion (HPT) experiments that have been carried out at low temperatures ($\sim -75^\circ\text{C}$) on the ternary alloys $\text{Cu}_{(85-x)}\text{Mo}_5\text{Ni}_x$ ($x = 0, 10$ and 20) alloys. Cu and Mo are extremely immiscible, while Ni has a relatively small heat of mixing with both Cu and Mo, and so the system corresponds roughly to $\text{Cu}_{(1-x)}\text{Co}_{(1-x)}\text{Cu}^*_{2x}$. The results of this study showed that the solubility of Mo in the Cu-rich phase increased with increasing concentration of Ni; this was attributed to an effective lowering of the heat of mixing of Mo in the Cu-Ni phase compared to that in pure Cu. The authors of that study reported, however, that the addition of 10% Ni to the Cu-Mo alloy increased the Mo solubility in Cu to a concentration $\sim 40\%$ greater than what could be attributed to the heat of mixing alone, while the increase in Mo solubility for the 20% Ni alloy was only $\sim 10\%$ greater than that predicted. We suggest here that a significant fraction of the excess enhancement of Mo solubility in the 10% Ni alloy was in fact due to Ni segregation to the Cu-Mo interface, accompanied by strengthening and an increased value of T_0 . For the 20% Ni alloy, this effect presumably saturates as most Cu-Mo bonds are replaced by Ni. Additional verification of this mechanism is clearly warranted. In this regard, it is known that carbon segregates to grain boundaries during HPT [81,82] during room-temperature processing, and it results in enhanced grain refinement and hardening. Presumably, carbon would also segregate to interphase boundary, and thus it would be of interest to determine if small amounts of C, or other highly mobile impurities, incorporated in two-phase alloys could influence phase stability by the mechanism suggested here.

4. Conclusions

The simulations presented here reveal that the properties of interfaces play a significant role in determining phase formation during steady state SPD. The mechanical properties of the co-existing phase are significant only in regard to their impact on the strength of the interface. The interface strength, however, can be controlled independently, to some extent, by appropriate impurity doping. These findings thus suggest a new strategy for developing nanocomposite alloys by severe plastic deformation with more controlled microstructures and properties. From a more fundamental perspective, the simulations show for the first time in nanocomposite alloys subjected to SPD, an equivalence between a system in a non-equilibrium steady-state and an effective system at thermal equilibrium without requiring thermally activated diffusion.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.actamat.2022.118333](https://doi.org/10.1016/j.actamat.2022.118333).

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